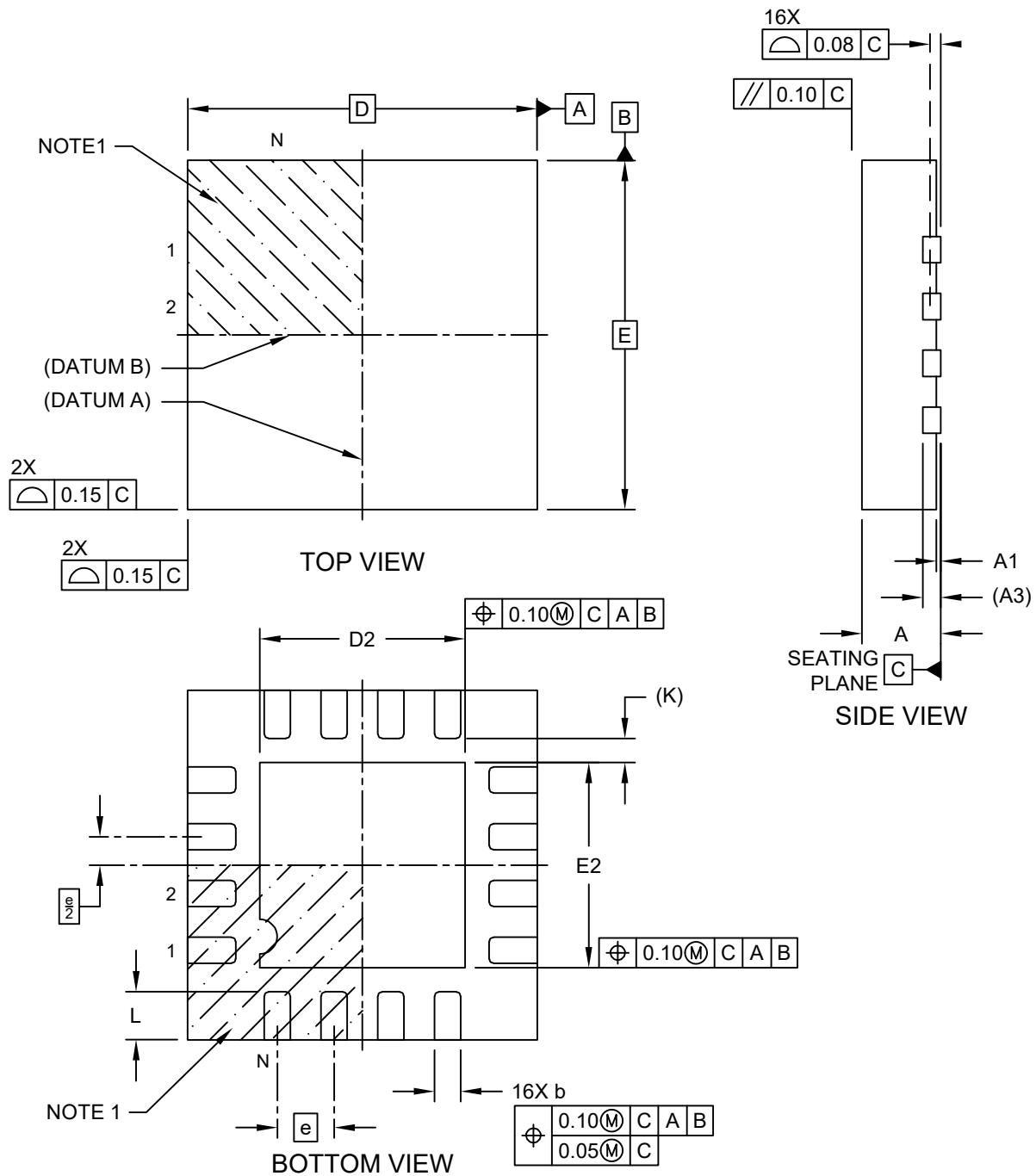


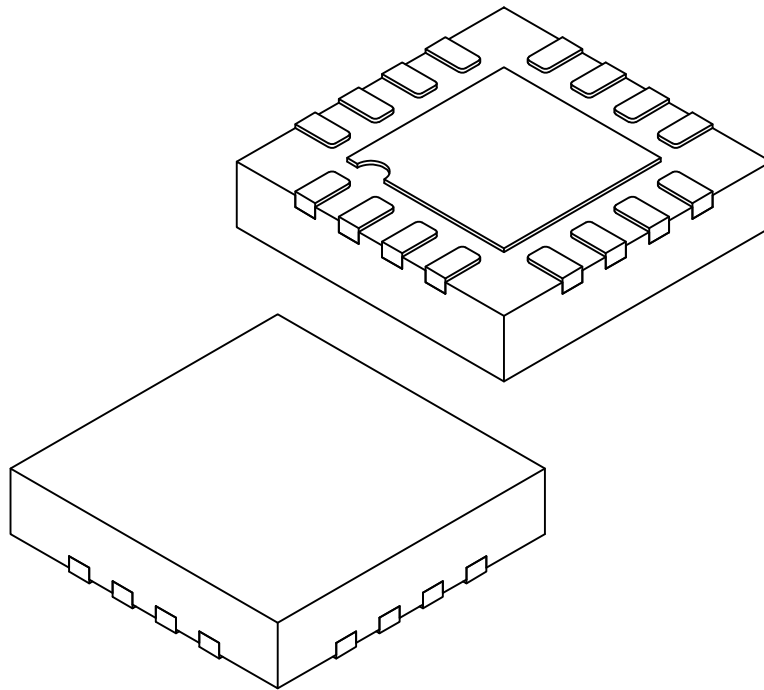
**16-Lead Very Thin Plastic Quad Flat, No Lead Package (NUA) - 4x4 mm Body [VQFN]  
With 2.35 mm Exposed Pad; Micrel Legacy Package MLF44Q-16L**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**16-Lead Very Thin Plastic Quad Flat, No Lead Package (NUA) - 4x4 mm Body [VQFN]  
With 2.35 mm Exposed Pad; Micrel Legacy Package MLF44Q-16L**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



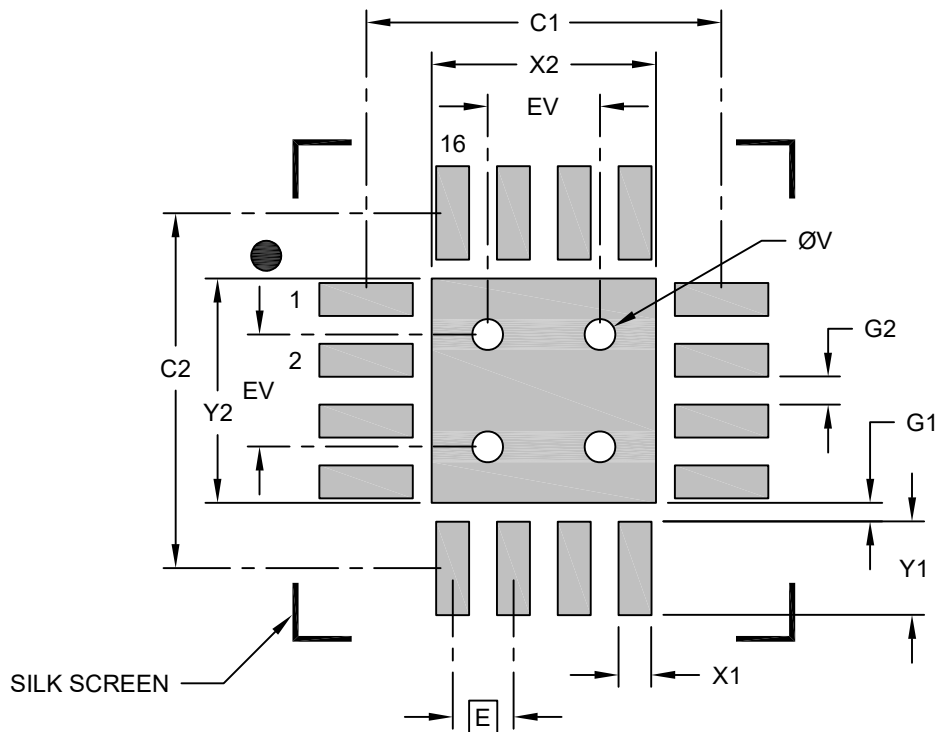
| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Number of Terminals     | N  | 16          |      |      |
| Pitch                   | e  | 0.65 BSC    |      |      |
| Overall Height          | A  | 0.80        | 0.85 | 0.90 |
| Standoff                | A1 | 0.00        | 0.02 | 0.05 |
| Terminal Thickness      | A3 | 0.203 REF   |      |      |
| Overall Length          | D  | 4.00 BSC    |      |      |
| Exposed Pad Length      | D2 | 2.30        | 2.35 | 2.40 |
| Overall Width           | E  | 4.00 BSC    |      |      |
| Exposed Pad Width       | E2 | 2.30        | 2.35 | 2.40 |
| Terminal Width          | b  | 0.25        | 0.30 | 0.35 |
| Terminal Length         | L  | 0.50        | 0.55 | 0.60 |
| Terminal-to-Exposed-Pad | K  | 0.28 REF    |      |      |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

**16-Lead Very Thin Plastic Quad Flat, No Lead Package (NUA) - 4x4 mm Body [VQFN]  
With 2.35 mm Exposed Pad; Micrel Legacy Package MLF44Q-16L**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.65 BSC    |      |      |
| Center Pad Width                 | X2 |             |      | 2.40 |
| Center Pad Length                | Y2 |             |      | 2.40 |
| Contact Pad Spacing              | C1 |             | 3.80 |      |
| Contact Pad Spacing              | C2 |             | 3.80 |      |
| Contact Pad Width (Xnn)          | X1 |             |      | 0.35 |
| Contact Pad Length (Xnn)         | Y1 |             |      | 1.00 |
| Contact Pad to Center Pad (Xnn)  | G1 | 0.20        |      |      |
| Contact Pad to Contact Pad (Xnn) | G2 | 0.30        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 0.20 |      |

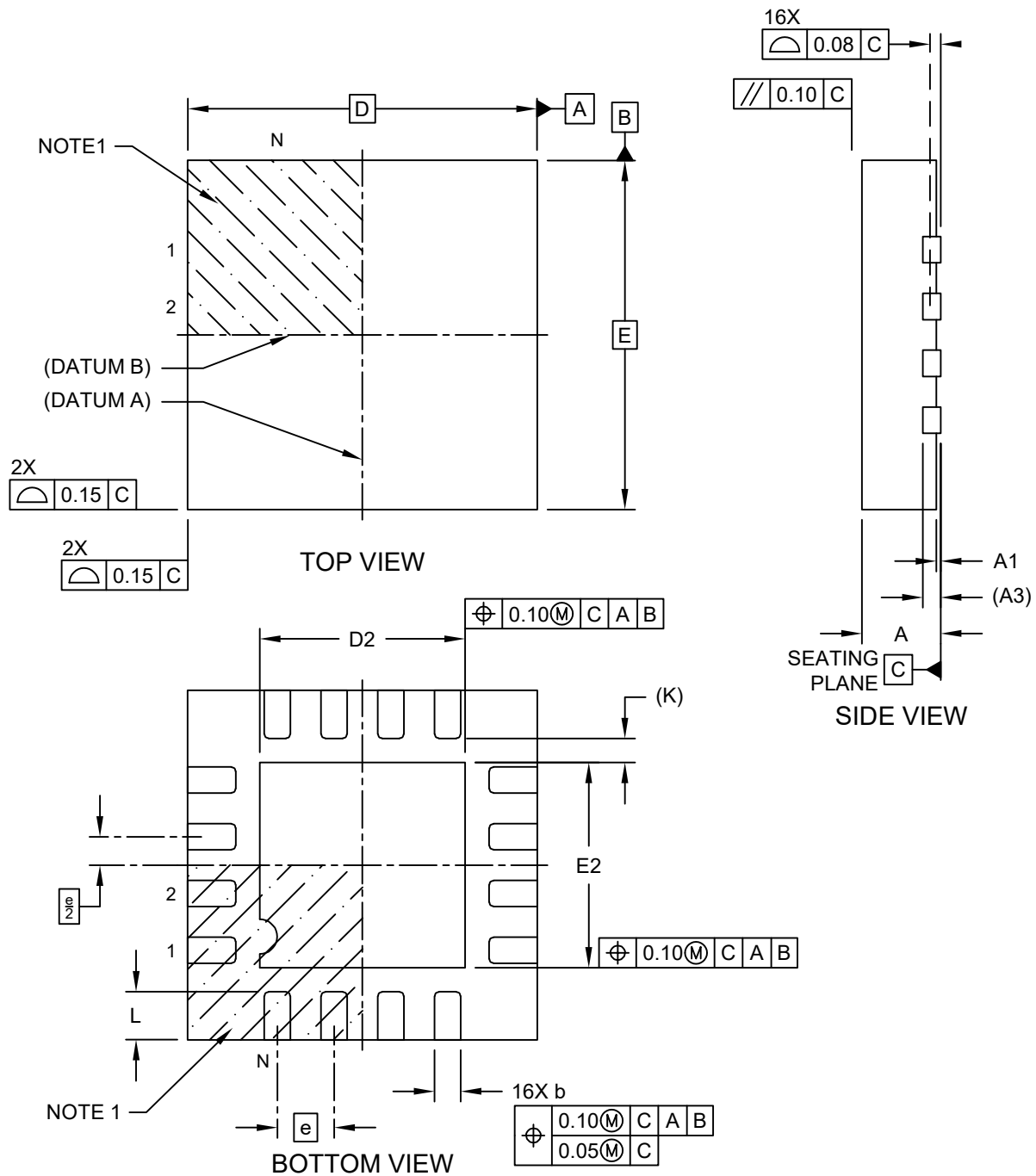
**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-3110-NUA Rev A

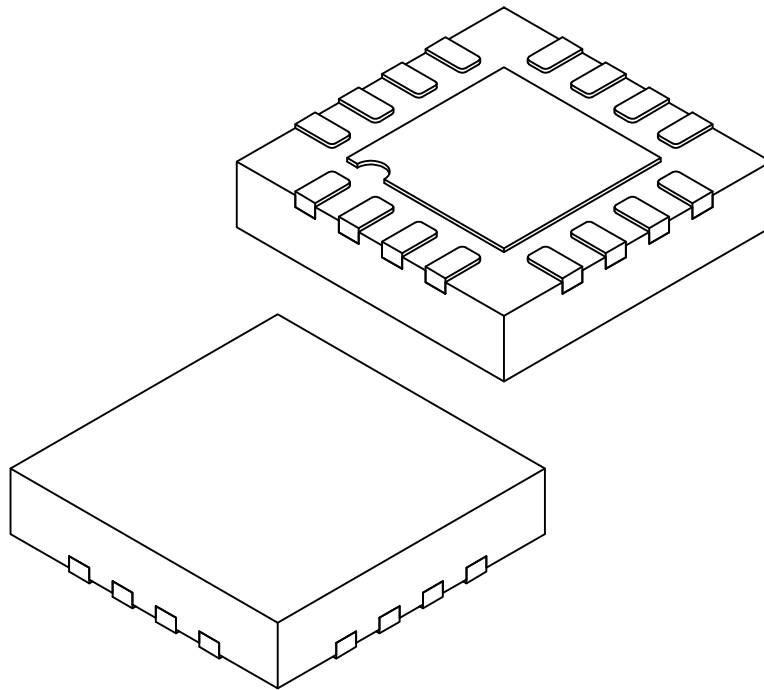
**16-Lead Very Thin Plastic Quad Flat, No Lead Package (NWA) - 4x4 mm Body [VQFN]  
With 2.35 mm Exposed Pad; Micrel Legacy Package MLF44Q-16L**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**16-Lead Very Thin Plastic Quad Flat, No Lead Package (NWA) - 4x4 mm Body [VQFN]  
With 2.35 mm Exposed Pad; Micrel Legacy Package MLF44Q-16L**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



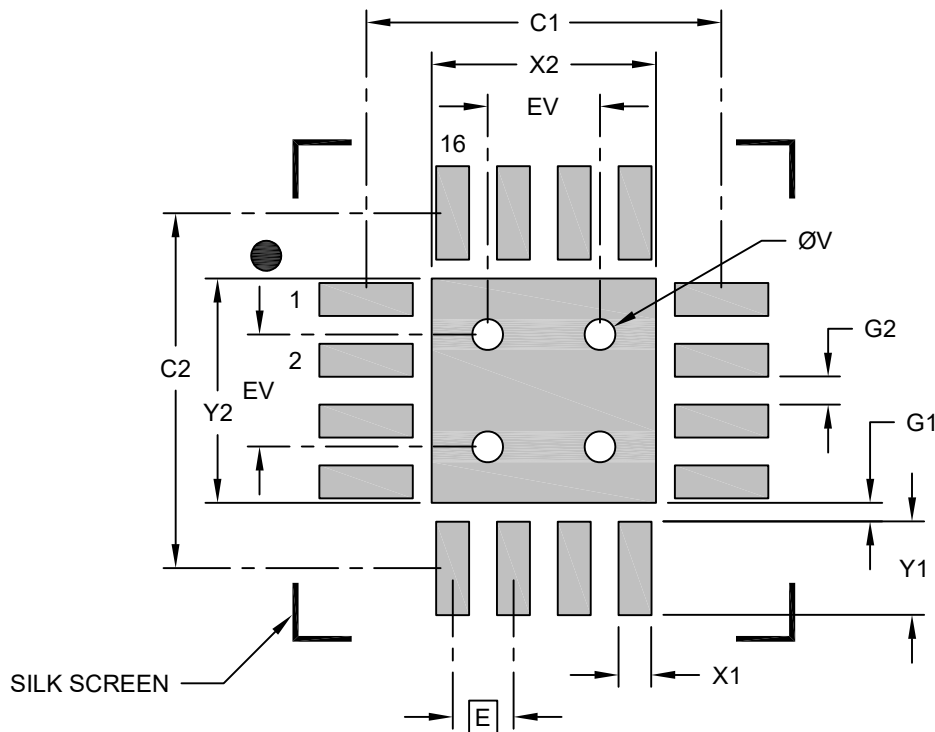
| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Number of Terminals     | N  | 16          |      |      |
| Pitch                   | e  | 0.65 BSC    |      |      |
| Overall Height          | A  | 0.80        | 0.85 | 0.90 |
| Standoff                | A1 | 0.00        | 0.02 | 0.05 |
| Terminal Thickness      | A3 | 0.203 REF   |      |      |
| Overall Length          | D  | 4.00 BSC    |      |      |
| Exposed Pad Length      | D2 | 2.30        | 2.35 | 2.40 |
| Overall Width           | E  | 4.00 BSC    |      |      |
| Exposed Pad Width       | E2 | 2.30        | 2.35 | 2.40 |
| Terminal Width          | b  | 0.25        | 0.30 | 0.35 |
| Terminal Length         | L  | 0.50        | 0.55 | 0.60 |
| Terminal-to-Exposed-Pad | K  | 0.28 REF    |      |      |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

**16-Lead Very Thin Plastic Quad Flat, No Lead Package (NWA) - 4x4 mm Body [VQFN]  
With 2.35 mm Exposed Pad; Micrel Legacy Package MLF44Q-16L**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**RECOMMENDED LAND PATTERN**

| Units                            |    | MILLIMETERS |      |      |
|----------------------------------|----|-------------|------|------|
| Dimension Limits                 |    | MIN         | NOM  | MAX  |
| Contact Pitch                    | E  | 0.65 BSC    |      |      |
| Center Pad Width                 | X2 |             |      | 2.40 |
| Center Pad Length                | Y2 |             |      | 2.40 |
| Contact Pad Spacing              | C1 |             | 3.80 |      |
| Contact Pad Spacing              | C2 |             | 3.80 |      |
| Contact Pad Width (Xnn)          | X1 |             |      | 0.35 |
| Contact Pad Length (Ynn)         | Y1 |             |      | 1.00 |
| Contact Pad to Center Pad (Xnn)  | G1 | 0.20        |      |      |
| Contact Pad to Contact Pad (Xnn) | G2 | 0.30        |      |      |
| Thermal Via Diameter             | V  |             | 0.33 |      |
| Thermal Via Pitch                | EV |             | 0.20 |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-3110-NWA Rev A